PANJAB UNIVERSITY CHANDIGARH- 160 014 (INDIA)

(Estted. under the Panjab Univerasity Act VII of 1947-enacted by the Govt. of India)

FACULTY OF SCIENCE

SYLLABI

FOR

ENVIRONMENT CONSERVATION (ELECTIVE) B.A./B.Sc. (General) 1st & 2nd YEAR

EXAMINATIONS 2012

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Environment Conservation
B.A/B.Sc. (General) 1st Year

Paper	Name	Time in Hrs	Theory Marks	Internal Assessment Marks	Total Marks
A	Enviroment & Forestry	3	65	10	75
B	Soil and water pollution	3	65	10	75
Practical					
A	Environment & Forestry	4	40	10	50
B	Soil and water pollution				
Total Marks					**200**

Note: The practical will include survey and its project reports carrying 10 marks, 30 marks will be allotted to laboratory practicals. The number of hours for theory and practical per week shall be 5 hours and 4 hours respectively.

There will be 9 questions in all, two each from section I to 1V. Question No. I (17 marks) will be short answer type & will cover the whole syllabus. Candidates will attempt five question in all, selecting one each (12 marks) from section I to IV and the first compulsory question.

Paper-A Environment and Forestry

Section-I

Environment and Natural Resources:

Environment:
Definition, Scope & importance of Environment. A brief introduction of Physical & biological environment.

Ecosystem:
Concept of Ecosystem, Biotic & abiotic components, food chain, food web trophic levels, types of ecosystems, terrestrial and aquatic. Biogeochemical cycles- nitrogen, carbon, phosphorous and sulphur cycle.

Section-II

Natural resources: Definition, type of natural resources (Renewable and nonrenewable natural resources) and the policies of their conservation. Energy resources; Fossil fuel, Alternative source of energy (Solar energy, wind power, geothermal energy, dung Energy and wood energy.

Section-III

Forestry:
Forestry: Concept, types of forests in India & the world. Deforestation, reforestation and afforestation. Major and minor forest products. Reasons of forest destruction and its impact on Environment. Community forestry, Farm forestry, Social forestry and Agro forestry.

Section-IV

Indoor environment:
Pollution of the in house environment pollutants in the offices, workplaces (School, Bus stand, College and Kitchens). Environmental problems linked to urban and rural lifestyle, Adulterants; Food adulterants (Wheat floor, milk, red chili powder, mustard oil, desi ghee, sweets, artificial sweetness, dyes, food allergens.)

Practical

Paper-A

Based on theory paper.

Laboratory work: Tests of food adulterants.

Study the forest products with help of charts and specimens.

Survey reports of indoor environmental pollutants and local forests.

A visit to forest to study different components of these ecosystems.

Paper-B Soil and water pollution

Section-I

Lithosphere:
Meaning of Soil profile, its components, types of soil, physical-chemical properties of soil.

Soil fertility:
Micro and macro nutrients, technique of testing soil sample. Methods of increasing soil fertility, merits and demerits of fertilizers, role of soil micro organisms.

Section-II

Degradation of soil; Soil erosion (Water and wind erosion) causes of effect of erosion.
Soil pollution: Different types of soil pollutants (Chemicals, Pesticides, Fertilizers & manure, discarded material. Pollution and control measures.

Section-III

Hydrosphere:
Major sources and uses of water. Over utilization of surface & ground water, floods, drought. Conflicts over water. Potable water, its characteristics. Water cycle (Global and Biological)

Section-IV

Water pollution:
Definition, Types & Sources of water pollution, its consequences and control measures. Different types of diseases due to water pollution. Treatment of waste water by green method (Root-zone technology), Marine pollution- a brief account.

Practical

Paper-B

Determination of soil pH

Use of Portable Kit

Determination of organic matter in soil

Determination of Ca CO_3 in soil.

Determination of available Nitrogen in soil.

Identification of different fertilizers.

Determination of pH of water.

Determination of organic carbon.

Determination of microbial carbon.

Determination of available N_2.

Determination of dissolved O_2, BOD.

Determination of hardness and alkalinity of water.

Survey reports of different water samples in local and adjoining areas and interpretation of water test report.

Books Recommended

Purochit S.S., Shammi Q.J. & Agarwal A.K	Text book of Environment Sciences, Sarsaswati pub., Jodhpur, 2007.
Dhaliwal, G.S & Kukul, S.S.	Essentials of Environmental Sciences, Kalyani Pub., Ludhiana,2008.
Chundawat, B.S. & Gutam, S.K	Text book of Agroforestry, Raju Primalai, New Delhi. 2006
Chandna, R.C.	Environmental Geography kalyani Pub. Ludhiana 2006
Sagreiya, K.P.	Forests and Forestry, National book truest, New Delhi.2007
Ralhon, P.K, Dhaliwal, G.S. & Avtar Singh	The Forests production and Management, Kalyani Pub., Ludhiana
Arora, S.	Fundamentals of Environmental Biology, Kalyani Pub., Ludhiana, 2008.
Dwivedi, A.P.	Forestry in India. Surya Publications, Dehradun. 1993.

Environment Conservation

B.A./B.Sc.(General) 2nd Year

Paper	Name	Time in Hrs	Theory Marks	Internal Assessment Marks	Total Marks
A	Air pollution and Climate change	3	65	10	75
B	Soil waste and Disaster Management	3	65	10	75
Practical					
A B	Air pollution and Climate change Solid waste and Disaster Management	4	40	10	50
Total Marks					**200**

Note: The practical will include survey and its project reports carrying 10 marks, 30 marks will be allotted to laboratory practical. The number of hours for theory and practical per week shall be 5 hours and 4 hours respectively.

There will be 9 questions in all, two each from section I to 1V. Question No. I (17 marks) will be short answer type & will cover the whole syllabus. Candidates will attempt five question in all, selecting one each (12 marks) from section I to IV and the first compulsory question.

Paper-A Air pollution and Climate Change

Section-I

Atmosphere: Composition of atmosphere, Air Pollution: Sources, Classification (Natural and anthropogenic sources, primary and secondary pollutants), Smoke, Smog, Particulate Matter and Acid rain. Effects of air pollution on human health, plants, animals and material. Air (Prevention and control of Pollution) Act, 1981.

Section- II

Green House Effect: Definition, Green house gases, Sources of GHE, Global Warming.
Stratospheric ozone layer depletion. Role of agriculture to green house effect (paddy and livestock, biomass burning) in GHE.

Section III

Noise pollution: Definition, Sources, Noise standards, Silence Zones, Consequences of noise pollution.

Radioactive Pollution: Definition**,** Sources, effect on human health

Mineral Resources: Types and Importance of minerals; Mineral extraction and environmental problems (limestone mining in Moussourie hills, coal mining in Jharkhand).

Section IV

Control: Measures and techniques to control air pollution in vehicles and industries, Green building concept, CDM (Clean Development Mechanism), Carbon sequestration. Control of noise pollution.

Practical

Paper-A

To monitor the level of Dust fall, SPM in surrounding atmosphere.

To monitor the level of NOx and SO2 in surrounding atmosphere.

To prepare a survey report on different sources causing air pollution in local area.

To prepare a survey report on noise pollution in your locality.

To visit, collect data and prepare report regarding air and water pollution form sugar mill, thermal plant, cement factory in the adjoining area like Nawanshahr, Ropar, and Hoshiarpur.

Paper-B

Solid Waste and Disaster Management

Section-I

Solid waste: Sources, nature and characteristics; categories of solid waste, Biodegradable and non biodegradable waste; Plastic nuisance. Solid waste disposal: Sanitary landfills, Composting, optimum conditions for composting, Pyrolysis, Incineration and Refuse Derived Fuels.
Fly Ash utilization, MSW Handling Rules, 2000.

Section-II

Disaster management: Natural disasters: floods, earthquakes, Tsunami, land slides, drought, forest fire, and Tropical cyclones.
Disaster management: pre-disaster phase, actual disaster phase, post- disaster phase.

Section-III

Bioenergy and waste water treatment: Biogas plant: Construction and working of biogas plants, its advantages and disadvantahges. Vermi-culture and Vermi-composting, Biofertilisers, Biofuels.
Waste water treatment (STP and ETP), A brief account of primary, secondary and tertiary treatment

Section-IV

Organic farming: Definition, Need of Organic Farming, Methods: Soil management, weed management, control of pests. Its advantages.
Organic Food, its benefits

Waste lands: Definition, Categories of waste lands, Desertification, ways of reclamation.

Practical

Paper-B

A visit to Composting/Vermicomposting Unit

To prepare a survey report of municipal waste of your city (Generation to disposal).

Classify the waste of your municipality waste (Source based and Waste based)

To visit any land fill site and prepare a tour report.

To prepare a list of herbicides / pesticides / insecticides commonly used in this Area

To make a list of various botanical pesticides available in the market

Assignments/project as assigned by the teacher

Books Recommended

Critchfield, Howard J. General Climatology. Second Edition, Prentice-Hall, Inc., 1982.

Houghton, J. Global Warming, The Complete Briefing. 3rd Edition, Cambridge Univ Press, 2006.

Hill, M. K. Understanding Environmental Pollution. Cambridge Uni Press. 2008.

Malsin, M. Global Warming, A Very Short Introduction. Oxford Univ. Press. 2009

Tchobanogloas, G. Integrated Solid Waste Management: Engineering, Principle and Management. McGraw Hill, USA. 1993.

Singh, H.M. Solid Waste Management. Alfa Publications, Delhi. 2010.

Bhatnagar R.K & Palta RK Earthworm vermin culture & vermi composting, Kalyani, Pub.2007.

Ranganathan L.S Vermi biotechnology from soil health to human health. Updesh purodit pub.Jodhpur,2006

Aggarwal, A. K. Water Pollution. APH Publications, New Delhi. 2008.

American Public Health Association (2003) Standard Methods for Examination of Water and Wastewater, APHA, New York.

Sengupta, B., John, S. & Abraham, T. Forth Estate, A Handbook on Air, Water and Land Pollution. IDMA Foundation for Sustainable Development and BR Publications Corporation, Delhi. 2006.

Aggarwal, A. K.	Noise Pollution. APH Publications, New Delhi. 2008.
Veeresh, G. K.	Organic Farming. Foundation Books, Delhi. 2006.
Gehlot, D.	Organic Farming: Componenets and Management. Agrobios, Jodhpur. 2010.
Singh, K. K. & Aleyer, L.	Disaster management: Manmade Disasters. APH Publications. 2010.
Dhunna, M.	Disater Management. Vayu Education of India, Delhi. 2009.
Sharma & Sharma	Natural Disasters. APH Puiblications. 2009.
Dara, S.S.	A Textbook of Environmental Chemistry. 2nd Edition. S. Chand & Company Ltd., Ram Nagar, New Delhi, 2010.

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